KK-P-91b
August 16, 1963

SUPERSEDING
Fed. Spec. KK-P-91a
January 17, 1951

FEDERAL SPECIFICATION

PALM, SEWING, (SAILMAKERS' AND SADDLERS')

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers leather sewing palms intended to be used as an aid in forcing the needle through heavy material when hand sewing such materials as canvas, leather, etc.

1.1.1 *Federal specification coverage.* Federal specifications do not include all varieties of the commodity indicated by the title of the specification, but cover only those generally used by the Federal Government.

1.2 Classification.

1.2.1 *Types.* The sewing palms shall be furnished in the following types, as specified (see 6.1):

Type I.—Roping.
Type II.—Seaming.

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification:

Federal Specifications:

L-P-378 — Plastic Film (Polyethylene Thin Gauge).
V-T-291—Thread, Linen.
KK-L-156 — Leather, Cattlehide, Harness, Vegetable-Tanned.
PPP-B-636—Box, Fiberboard.
PPP-T-60 — Tape; Pressure-Sensitive Adhesive, Waterproof, for Packaging and Sealing.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).
Fed. Std. No. 751—Stitches, Seams and Stitchings.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, and Auburn, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:56Z
Check the source to verify that this is the current version before use.

Military Specification:

MIL-R-1978—Rawhide, Hard.

Military Standards:

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129—Marking for Shipment and Storage.

Drawing—Quartermaster Corps:

2-1-627—Palm, Sewing (Sailmakers' and Saddlers'). (See fig. 3.)

(Copies of Military Specifications, Standards and Drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.

3.1.1 *Leather.* The leather used in the manufacture of the palms shall conform to type II of KK-L-156 and the color shall be russet.

3.1.1.1 *Rawhide.* The rawhide for the palms shall conform to type a or b of MIL-R-1978.

3.1.2 *Thread.*

3.1.2.1 *Machine stitching.* The thread used in machine stitching the palms shall be linen conforming to type B, class 1 or 2, 4 ply of V-T-291.

3.1.2.2 *Hand stitching.* The thread used in hand stitching the palms shall be linen conforming to type C, class 1 or 2, No. 10 of V-T-291.

3.1.3 *Metal.* The metal for the thimble shall be a good grade of malleable iron.

3.2 Construction.

3.2.1 *General design.* Unless otherwise specified (see 6.2), the sewing palms shall be made for use on the right hand. Types I and II palms shall conform to the general design as shown in figures 1 and 2. They shall be so designed and formed as to fit the palm of the hand to be comfortable to the wearer and to facilitate the function for which they are intended. The thimble shall be located at the proper angle and so positioned that during sewing operation, the needle, when held between the thumb and index finger, will be given proper support.

3.2.2 *Main body construction.*

3.2.2.1 *Type I, roping palms.* The main body of the type I palm shall be made from leather (see 3.1.1) 10/64 to 12/64 inch in thickness. The grain side of the leather shall be outermost on the palm. A reinforcing strip of leather (see 3.2.1), 8/64 to 9/64 inch thick by 4-1/2 inches long, to which has been riveted or stapled a thimble, shall be placed on the grain side of the body section, extending back from the thumb opening, as a reinforcement for the thimble. The body shall be faced with a rawhide strip (see 3.1.1.1), 8/64 to 10/64 inches thick, extending from the end of the body leather across the palm and around the thumb hole. The thumb hole shall be circular or slightly oval shape so as to fit the operator's thumb and be comfortable to the wearer. The thimble, located properly beside the thumb hole, shall be securely held in position and completely encased by the rawhide facing except for the circular portion of the thimble exposing the indentations. This opening shall be 3/8 inch plus or minus 1/16 inch. The thumb portion shall be flanged (rolled) at the top edge in a suitable manner so that the sewing thread may be wrapped around that portion of the palm to facilitate drawing the stiching tight during heavy sewing operations. The type I palm shall conform to the dimensions specified in table I.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:56Z
Check the source to verify that this is the current version before use.

TABLE I.—*Dimensions for type I*

	Minimum inches	Maximum inches
Width of body at base of thimble	2	2-1/4
Opening for hand:		
Top width	1-3/8	1-5/8
Bottom width	3	3-3/8
Top length	4	4-1/2
Bottom length	4-1/2	5
Diameter of thumb hole	1-1/8	1-3/8
Length of rawhide from thumb hole	8	8-1/2
Width of leather at ends where joined	1/2	5/8

3.2.2.2 *Type II, seaming palms.* The type II palms shall be made from leather components of the same thickness as specified for the type I palms. The type II palms shall conform to the patterns as outlined in Quartermaster Corps Drawing 2-1-627 (fig. 3). The *patterns contain the positioning, holes, and the stitch design for the type II palms. The drawing also contains the details of the thimble, contours, measurements, and the underclaws for fastening. When positioned in the two holes in the leather overlay (reinforcing strip), the thimble assumes the proper positioning. Staples, when used for positioning of the body overlays, shall be removed after stitching. The thumb portion of the type II palm need not be flanged.

* A tolerance of plus or minus 1/8 inch shall be allowed for dimension specified in the drawing.

3.2.3 *Thimble plate.* The thimble plate for the type I and type II palms shall be made of metal as specified in 3.1.3. The design measurements, contours, and the face indenture shall be as detailed in Drawing 2-1-627 (fig. 3).

3.2.4 *Stitching.* A line of machine stitching shall extend entirely around the rawhide covering, approximately 3/16 inch from the edge and around the thimble base. The stitch design shall follow the design as out-

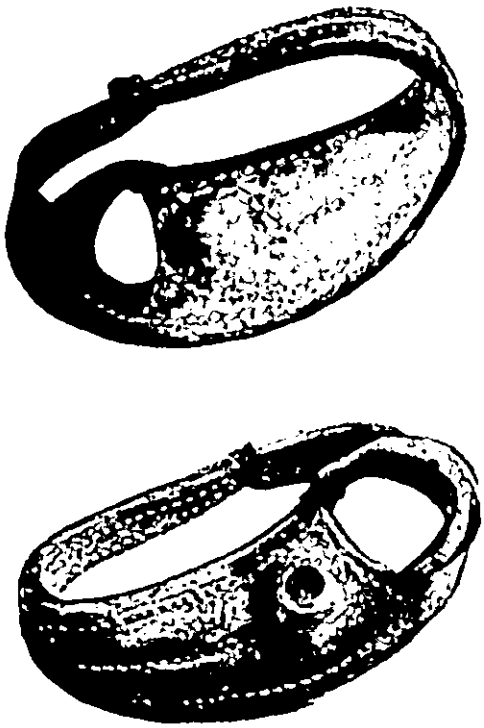


FIGURE 1.—Type I, roping palm.

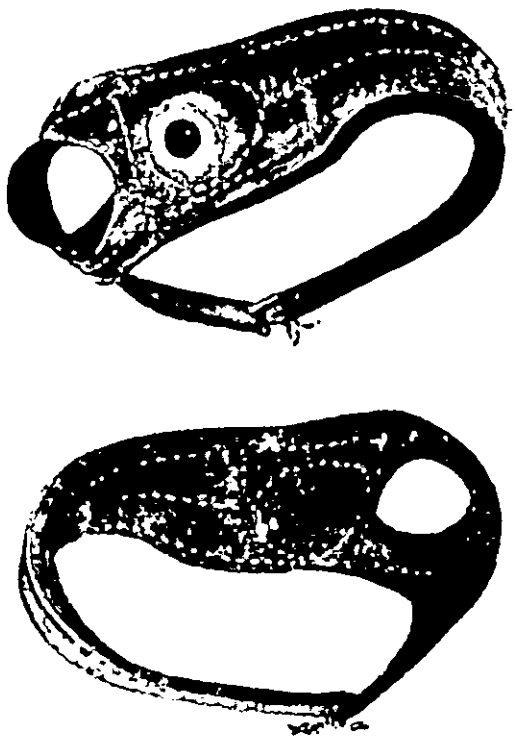


FIGURE 2.—Type II, seaming palm.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:56Z
Check the source to verify that this is the current version before use.

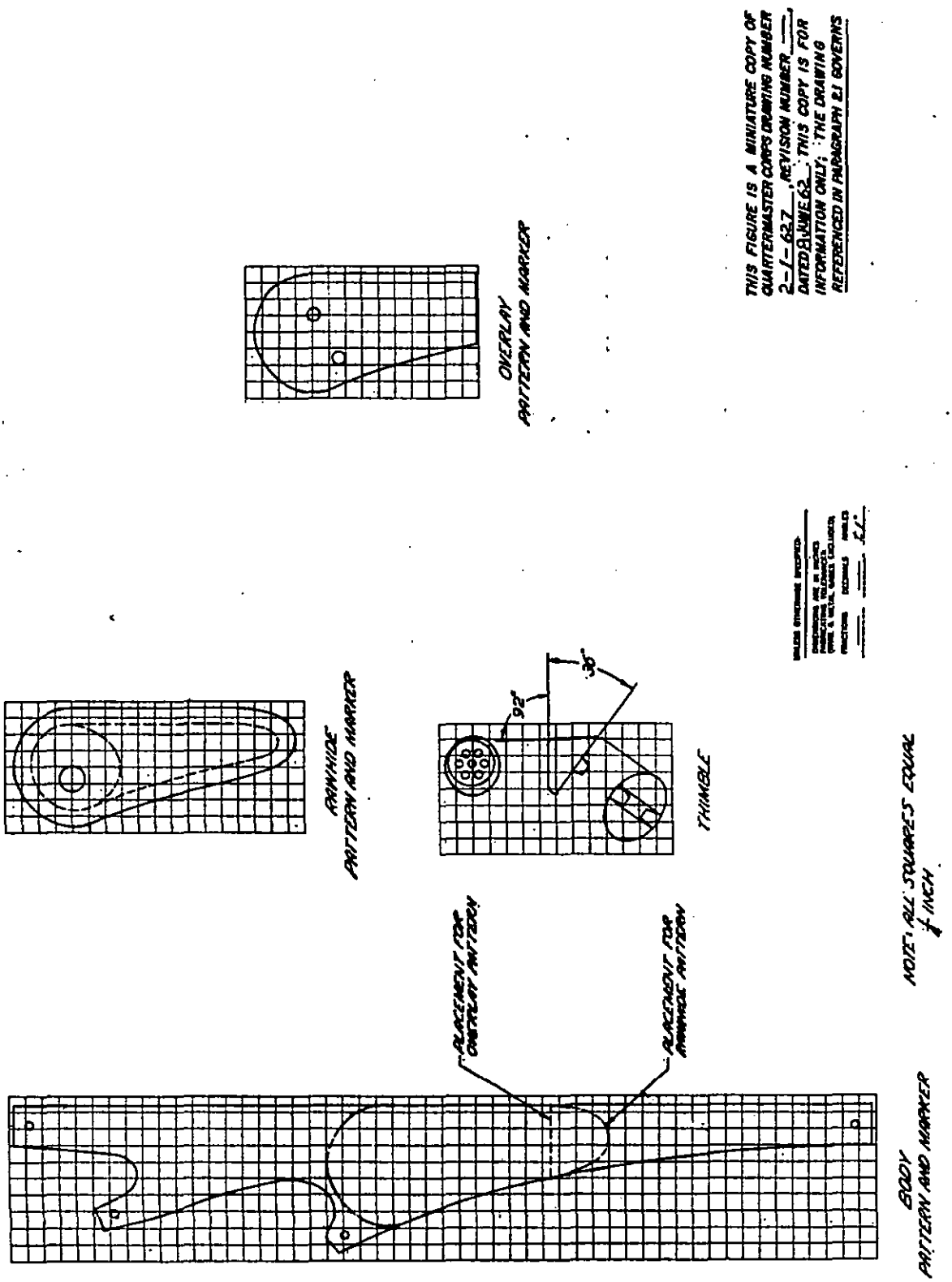


FIGURE 3.—Palm, sewing, (sailmakers' and saddlers').

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:56Z
Check the source to verify that this is the current version before use.

lined in Drawing 2-1-627, and may include an extra row of stitching through the center portion of the palm, beginning at the base of the thimble, as outlined in figures 1 and 2. The stitching shall conform to stitch type 301 of Fed. Std. No. 751 and shall be 5 to 7 stitches per inch. The stitches shall extend entirely through the layers of leather and rawhide (when used). The lock of the stitch shall be embedded in the leather. The ends of the leather shall be joined by hand sewing, using linen thread (see 3.1.2.2), to form abutting seams at the thumb and body end areas.

3.3 Finish. All edges of the palm shall be smooth and rounded. The rawhide portion of the fabricated palm, after forming to shape, shall be coated with shellac as a final operation.

3.4 Marking for identification. The grain side on the back area of the palm body shall be marked with the manufacturer's name or easily identifiable trademark. The marking shall be permanently impressed into the grain side of the leather without cutting through at any point.

3.5 Workmanship. The finished palms shall conform to the quality and grade of product established by this specification. The occurrence of defects shall not exceed the applicable acceptable quality levels.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Inspection for acceptance. Inspection shall be in accordance with the provisions of MIL-STD-105, except where otherwise specified herein.

4.2.1 *Inspection of materials and components.* The supplier is responsible for insuring that materials and components used were manufactured and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified herein or, if none, in accordance with the requirements of this specification.

4.2.2 *Examination of the end item.* The defects found during the examination of the end item shall be classified in accordance with 4.2.2.1 and 4.2.2.2. The sample unit for these examinations shall be one completely fabricated palm of each type offered for delivery at one time. The lot size shall be expressed in units of one palm. The inspection level and acceptable quality level for these examinations shall be in accordance with 4.2.2.3.

4.2.2.1 *Visual examination.* The palm shall be examined for defects in finish, design, material, workmanship and marking.

TABLE II.—*Classification of defects*

Examine	Defect
Leather	Not color specified. Any hole, cut, or rupture. Flanky, spongy, or stretchy leather.
Construction and workmanship	Any component missing, misplaced, or malformed. Staples not removed. Rawhide not coated with shellac. Any edge not smooth or rounded. Thumb portion not flanged (rolled) —type 1 only.
Seams and stitching	Not stitch type specified. Loose stitch tension resulting in a loosely secured seam or locks exposed on surface of leather. Tight stitch tension. Four or less stitches per inch. Eight or more stitches per inch. Any open seam.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:56Z
Check the source to verify that this is the current version before use.

TABLE II.—*Classification of defects (cont'd)*

Examine	Defect
Seams and stitching (cont'd)	Note: A seam shall be classified as open when one or more stitches joining a seam are broken or when two or more consecutive skipped stitches or runoffs occur.
Marking	Missing, incomplete, incorrect, not in proper location.
Markings (exterior and interior)	Omitted, incorrect, illegible, of improper size, location, sequence, or method of application.
Materials	Any component missing. Any component damaged, affecting serviceability.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling, bulging or distortion of containers, improper taping, open and noncontinuous heat-sealed seams and closure of polythylene bags.
Content	Number per container is more or less than required.

4.2.2.2 *Dimensional examination.* The palm shall be examined for defects in dimensions. Any dimension that is not within the established tolerance shall be classified as a defect.

4.2.2.3 *Inspection levels and acceptable quality levels.* The inspection levels and acceptable quality levels (AQLs) expressed in defects per 100 units shall be as follows:

For defects applicable to	Inspection level	AQL's
4.2.2.1	II	6.5
4.2.2.2	L-6	4.0

4.2.3 *Examination of preparation for delivery requirements.* An examination shall be made to determine that packaging, packing, and marking requirements of section 5 of this specification are complied with. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully packed, selected just prior to the closing operation. Defects of closure listed below shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of containers in the inspection lot. The inspection level shall be L4 and the AQL shall be 4.0 defects per hundred units.

5. PREPARATION FOR DELIVERY

(**Civil agencies.** Fed. Std. No. 102 should be referred to for definitions of the various levels of packaging protection for supplies and equipment.)

5.1 **Packaging.** Packaging shall be level A or C, as specified (see 6.1).

5.1.1 *Level A.* Each sewing palm shall be placed in a snug-fitting polyethylene film bag. The polyethylene flexible film shall be 0.0015 inch thick conforming to type I, grade A, B, or C, finish 1 or 2 of L-P-378. All seams and closure shall be effected by heat sealing. Prior to or during the heat sealing closure operation, excess air within the bag shall be expelled.

5.1.2 *Level C.* Sewing palms shall be packaged in accordance with the industry's practice.

5.2 **Packing.** Packing shall be level A, B, or C, as specified (see 6.1).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:56Z
Check the source to verify that this is the current version before use.

5.2.1 *Level A.* One hundred sewing palms of one type only, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to class 2, V3c or V3s, style RSC of PPP-B-636. Each fiberboard shipping container shall be closed and reinforced with flat steel strapping or tape banding in accordance with the appendix of the container specification. In addition, each shipping container shall have all seams, corners, and manufacturer's joint sealed by the application of a 3 inch minimum width olive drab color cloth tape conforming to type III, class 1 of PPP-T-60.

5.2.2 *Level B.* One hundred sewing palms of one type only, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to type I or II, class 1, style RSC of PPP-B-636. The minimum bursting strength of the fiberboard shall be 275 p.s.i. All flaps of the fiberboard shipping containers shall be securely sealed with an adhesive commercially used for the specific product being packed by application throughout the entire area of contact between the flaps, or by a combination of metal stitching the bottom flaps and sealing the top flaps with adhesive.

5.2.3 *Level C.* Sewing palms, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with rules or regulations of carriers applicable to the mode of transportation.

5.3 Marking.

5.3.1 *Civil agencies.* In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 *Military requirements.* In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

5.3.3 *Polyethylene-bagged packages.* Polyethylene-bagged packages shall have the required information legibly printed or stamped in black directly on the bag across the center face or on a white paper label inserted within the bag so as to permit ready identification. The bag or label shall indicate the following information:

STOCK NUMBER
NOMENCLATURE
TYPE

6. NOTES

6.1 Ordering data. Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.
(*b*) Type required (see 1.2).
(*c*) When left-handed palms are required (see 3.2.1).
(*d*) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
(*e*) Marking, if different (see 5.3).

6.2 Transportation description. Transportation descriptions and minimum weights applicable to this commodity are:

Rail:

Leather goods, not otherwise indexed by name.
Carload minimum weight 20,000 pounds, subject to Rule 34, Uniform Freight Classification.

Motor:

Leather goods, not otherwise indexed.
Truckload minimum weight 20,000 pounds, subject to Rule 115, National Motor Freight Classification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:56Z
Check the source to verify that this is the current version before use.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY CUSTODIANS:

Army—GL

Navy—Sh

Air Force—MOA

Copies of this specification may be purchased for 5 cents each.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:56Z
Check the source to verify that this is the current version before use.